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Failure Prevention and Recovery Techniques in Cloud-Based Systems: A Systematic Review

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Abstract

Cloud computing systems help to maintain massive distributed applications as well as data-intensive services in various fields. Nonetheless, the distributed and complex nature of the cloud infrastructures render them susceptible to a number of system failures, such as hardware failures, software failures, network failures, virtualization failures, and ineffective resource allocation. Such failures may have a great impact on the availability of services, the reliability of the system, and the general performance. This paper aims to provide a systematic review on the topic of preventing failures and recovery methods in cloud systems in a systematic way to offer an organized view of the existing methods of promoting network reliability. Based on the systematic search strategy in the ScienceDirect database, the sources related to interventions that were published in the last ten years (2015 to 2025) were searched and filtered according to the pre-established inclusion and exclusion criteria, which have led to the identification of a final dataset of 68 studies. The review evaluates proactive failure prevention methods including predictive monitoring, machine learning-based anomaly detection, redundancy plan, load balancing plan and intelligent resource scheduling. Moreover, recovery strategies such as checkpointing, replication, failover systems, container migration and disaster recovery systems are discussed. The authors find that proactive monitoring strategies in combination with effective recovery strategies are highly effective in enhancing the resilience and operational continuity of distributed cloud infrastructures. The paper also finds significant research issues associated with scalability, security risk, real-time monitoring, and multi-cloud reliability. Such results indicate that there is a necessity of coherent resilience frameworks which integrate intelligent monitoring, intelligent fault-tolerance schemes, and automatic recovery systems. The paper offers an organized insight into the approaches of preventing failures and recovery mechanisms of cloud computing infrastructure and the fact that smart and adaptable resilience models are necessary.

Keywords: Cloud computing; Failure prevention; Fault tolerance; Failure recovery; Predictive monitoring

1. Introduction

Cloud computing is now a paradigm of current technology that allows organizations to access and deploy scalable computing resources, storage system and distributed processes platforms using the internet. The blistering growth of cloud infrastructures in the last ten years has very tangibly changed the way contemporary applications are implemented and managed in all sectors, including healthcare, transportation, smart energy systems, and digital governance (Abdi et al., 2024; Ahmed et al., 2025; Abdullayeva, 2023). Combining cloud computing with the new technologies such as Internet of Things (IoT), artificial intelligence, and edge computing has further enhanced the functionalities of distributed computing systems allowing one to conduct real-time analytics, massive data processing, and smart automation (Pentyala, 2024; Ahmed et al., 2025; Ibitoye, 2018). Besides, recent research has become keen on the increasing relevance of cloud-enabled architectures to the support of large-scale digital ecosystems and intricate

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cyber-physical systems (Panful et al., 2025; Shandilya et al., 2024). Notwithstanding these benefits, there are also major challenges of cloud platform reliability, scalability, and operational resilience with the growing dependency on cloud platforms.

With the continuous evolution of the cloud infrastructures, the complexity of the infrastructures grows because of the interplay of various distributed components such as the virtual machines, containers, microservices, storage systems, and network infrastructures. These types of architectures can introduce additional risks of system failures, which can occur due to the failure of hardware, bugs in software, resources allocation error, or network outages. In cloud-based applications, failures may cause a major effect on the performance of the system, the data integrity and its availability (Garrahan et al., 2018; Achary, 2021). Furthermore, current cloud systems become increasingly based on multi-cloud implementations and containerized architectures that demand complex orchestration and monitoring systems to ensure the stability and reliability (Gupta, 2024; Torres Ponce, 2025). It has also been found that the growing involvement of IoT systems and distributed edge infrastructures enhances the reliability problem since the numbers of heterogeneous computing environments and dynamic workloads continue to rise (Lampropoulos et al., 2024; Ibitoye, 2018; Pentyala, 2024). This has made the provision of dependable and fault-tolerant cloud systems one of the key studies in the field of distributed systems engineering.

In order to deal with these issues, scholars have recommended so many strategies that would enhance the reliability and durability of cloud computing environments. Overall, these strategies may be divided into failure prevention strategies and failure recovery models. The prevention of failures strategies inculcates the proactive identification of possible system anomalies and the prevention of risks that may result in failures. Such methods usually entail prediction-based monitoring, intelligent resource management, anomaly detection, and automated system control systems, aimed at enhancing the stability of a system (Pentyala, 2024; Ahmed et al., 2025). The further companies have introduced intelligent monitoring systems that can identify an abnormal behavior of the system and avoid failures in distributed computing infrastructures (Ofili et al., 2024; Ahmed et al., 2025). Also, there are new architectures like edge computing, distributed intelligence systems which are helping in enhancing responsiveness of the system and the operational risks when using large-scale clouds (Gupta, 2024; Andreyev, 2025).

Even though prevention methods can be effective, failure of big-scale distributed systems cannot be avoided. Hence, there must be efficient failure recovery systems that will restore the functionality of the system and the delivery of the service. The common way to recover involved checkpointing, replication, failover, container migration, and disaster recovery framework strategies that aim to minimize the service disruption and provide data access in the event of a system failure (Rana & Giri, 2026; Thumala, 2020). Some articles have already highlighted the significance of using both proactive and reactive strategies to improve distributed computing environments resilience (Welsh & Benkhelifa, 2020; Mukwevho & Celik, 2018). Nevertheless, the current studies tend to concentrate on technologies or areas of application, including IoT security, blockchain, or distributed architecture instead of offering an overview of failure prevention and recovery methods in the cloud-based systems (Abdi et al., 2024; Sahar Saeed et al., 2025). Therefore, the study of the literature that is structured and systematic to organize the available knowledge, classify the existing techniques, and define the new research questions in the field has not been carried out yet. Thus, the paper is a systematic literature review to analyze the available literature on preventing failure and recovering methods in cloud-based systems, typify the extant methods, and establish gaps and research thrusts in the future that can enhance the reliability and resilience of the modern cloud systems infrastructure.

2. Systematic Review Methodology

A systematic literature review (SLR) is a clear and organized procedure of finding, choosing, and evaluating pertinent literature investigations in a particular area. As a contrast to classic literature reviews, SLRs use well-defined search strategies, screening processes, and data-generating patterns to reduce bias and guarantee the replicability of their findings. Such a methodological approach allows researchers to systematize the current knowledge base and determine the trends and gaps in the research, as well as new prospects in the field of research (Welsh & Benkhelifa, 2020; Abdi et al., 2024). Systematic reviews have been extensively applied in the study of technological advances, reliability issues, and the introduction of new architectural models in the distributed computing environment in the context of cloud computing research (Ahmed et al., 2025; Mukwevho & Celik, 2018).

In this work, the systematic review approach will be used to examine the current literature on the topic of failure prevention and recovery approaches in cloud-based systems. A review process is based on the structured workflow which has four primary steps: literature search, study selection, data extraction, and analysis. The methodological process was developed in a manner that would provide transparency, reproducibility and at the same time provide a comprehensive coverage of the relevant studies published during the specified search period. The review process is in

Table 4 Data Extraction Framework

Research Category	Number of Studies	Key Techniques Identified	Example Studies	Evaluation Approaches	Key Contribution to Cloud Reliability
Cloud security and privacy mechanisms	16	Blockchain security, encryption frameworks, authentication mechanisms	Abdullayeva (2023); Czuryk (2023); George et al. (2024)	Literature reviews, surveys, comparative analysis	Enhancing cloud security to prevent system vulnerabilities and cyber-attacks
Blockchain-based reliability frameworks	10	Distributed ledger technologies, consensus algorithms, decentralized verification	Abdi et al. (2024); Sahar Saeed et al. (2025); Welsh & Benkhelifa (2020)	Systematic reviews and analytical evaluations	Improving trust, data integrity, and reliability in distributed cloud systems
Edge and distributed computing resilience	12	Edge intelligence, computation offloading, distributed resource allocation	Pentyala (2024); Andreyev (2025); Gupta (2024)	Analytical studies and surveys	Reducing latency and improving fault tolerance in distributed cloud infrastructures
Artificial intelligence and machine learning techniques	9	Federated learning, predictive analytics, AI-driven monitoring	Pentyala (2024); Jha (2025); Ofili et al. (2024)	Systematic reviews and conceptual frameworks	Intelligent fault detection and predictive failure prevention
IoT and cyber-physical system reliability	11	IoT-cloud integration, digital twins, monitoring frameworks	Ibitoye (2018); Lampropoulos et al. (2024);	Systematic mapping and literature synthesis	Enhancing monitoring and predictive maintenance in distributed systems
Cloud monitoring and DevOps management	5	Monitoring tools, microservice observability frameworks	Jha (2025); Torres Ponce (2025); Gupta (2024)	Grey literature reviews and empirical analysis	Early detection of system anomalies in cloud environments

3. Overview of Cloud System Failures

This section provides a descriptive overview of cloud system failures based on the synthesized literature and serves as a foundation for the subsequent analysis. It focuses on identifying and categorizing the major types of failures reported in cloud computing environments without yet evaluating specific techniques. The findings and critical analysis of failure prevention and recovery techniques are presented in Sections 4, 5, and 6, where the results of the systematic review are discussed in detail.

Cloud computing services are systems of distributed data centers, virtualization platforms, and networked resources, which together provide the computing services and digital applications of large scale. These infrastructures assist the organizations with processing huge amounts of data and implementing scalable applications in geographically dispersed settings (Abdi et al., 2024; Ahmed et al., 2025). Nevertheless, the dispersed characteristic of the cloud systems provides numerous possible areas of failure such as hardware errors, software bugs, network outages, and ineffective allocation systems (Garraghan et al., 2018; Achary, 2021). With more technologies, like IoT systems, edge computing structures, and artificial intelligence-based services becoming part of a cloud environment, interactions with a system become highly complex, which makes it more likely that system processes suffer disruption (Ibitoye, 2018; Pentyala, 2024; Lampropoulos et al., 2024). As a result, it is necessary to learn about the causes and consequences of failures in cloud infrastructures that can be used to design efficient prevention and recovery strategies (Welsh & Benkhelifa, 2020; Mukwevho & Celik, 2018).

will not require starting a whole process again once the disruptions take place (Rana & Giri, 2026; Thumala, 2020). Checkpointing systems are typically provided by virtual machine snapshots or container checkpoint systems, which store system state(s) at execution (Gupta, 2024; Szorc, 2023). Replication is another significant recovery mechanism, during which critical data and services are replicated on multiple servers or storage systems to be available in the event of unavailability because of a system component failure (Welsh & Benkhelifa, 2020; Achary, 2021). Replication strategies enhance the reliability of the system since the failed components can be immediately replaced by backup nodes but can raise storage and infrastructure costs (Nikolovski et al., 2025; Gupta, 2024).

Along with checkpointing and replication, cloud infrastructures are based on failover systems, container migration approaches, and disaster recovery systems to ensure that they cannot encounter disruptions in service delivery in case of failures. Failover mechanisms automatically direct the workload of failed servers to backup computers so that applications can keep running without causing any major disruption of the service (Andreyev, 2025; Kenna, 2024). In like manner, container migration enables the migration of running applications between computing nodes in case of failures or performance concerns, thus, enabling operational continuity within containerized cloud environments (Gupta, 2024; Torres Ponce, 2025). Disaster recovery systems also facilitate scaling recovery in that they sustain backup systems and recovery procedures to allow cloud platforms to address significant failures including data center failures, data breaches or natural disasters (Ahmed et al., 2025; Filani & Opoku, 2025). All these recovery mechanisms guarantee that cloud systems will be resilient and be able to bring back the services fast in such dynamic distributed computing environments (Mukwevho & Celik, 2018; Adjei et al., 2026; Obidiagha & Antwi, 2026).

Table 6 Recovery Mechanisms in Cloud Systems

Study	Recovery Technique	Implementation	Performance Impact
Rana & Giri (2026)	Checkpointing	Virtual machine snapshot	Fast recovery with minimal downtime
Thumala (2020)	Checkpointing	System state saving during computation	Reduces process restart time
Welsh & Benkhelifa (2020)	Replication	Distributed data duplication	Improved reliability but increased storage cost
Achary (2021)	Replication	Distributed secure data storage	Enhances availability and fault tolerance
Andreyev (2025)	Failover mechanism	Backup server activation	Reduces service interruption
Kenna (2024)	Failover mechanism	Dynamic resource reassignment	Maintains system continuity
Gupta (2024)	Container migration	Container relocation across nodes	Maintains service continuity
Torres Ponce (2025)	Container migration	Cloud-based data exchange framework	Improves workload recovery
Sahar Saeed et al. (2025)	Disaster recovery system	Backup infrastructure and recovery protocols	Enables recovery from large-scale disruptions
Abdi et al. (2024)	Disaster recovery system	Secure distributed healthcare recovery frameworks	Improves system resilience

6. Comparative Analysis of Prevention vs Recovery Techniques

Failure prevention and failure recovery methods are the two complementary approaches to increasing reliability and resilience within the cloud computing environment. Prevention methods are based on the fact of detecting the possible faults of the system in its operation before the event, whereas recovery methods are aimed at restoring the services after the event. The proactive prevention systems that include predictive monitoring, anomaly detection, and smart resource allocation are trying to minimize the disruption chances by continuously examining the system behavior and identifying anomalies at the early stages (Welsh & Benkhelifa, 2020; Mukwevho & Celik, 2018). By contrast, checkpointing, replication, and failover mechanisms are recovery mechanisms that are aimed at reinstating the functionality of the system after failures have taken place, so that services are not affected by the dysfunction of the system (Abdi et al., 2024; Thumala, 2020). With the increased complexity of cloud infrastructures, the two approaches

are being merged to offer a holistic approach to failure management within distributed computing setup (Gupta, 2024; Rana & Giri, 2026).

The first significant disparity between the approaches of prevention and recovery methods is associated with the time of operations and the strategy. The prevention methods are proactive, so they keep track of the system behavior and determine some possible risks before failures have taken place, which would decrease the chances of system disruptions. Machine learning-based anomaly detection and predictive monitoring frameworks are all types of technologies that enable cloud systems to identify abnormal trends in system performance and take action before the faults develop into significant failures (Ahmed et al., 2025; Ofili et al., 2024). Recovery techniques, on the other hand, are reactive in which services are restored after failure has already happened. Checkpointing, container migration, and replication are the mechanisms that allow cloud infrastructures to resume their operations within a short period of time and reduce service outages (Rana & Giri, 2026; Torres Ponce, 2025). The two methods are thus vital because prevention provides the opportunity to minimize the occurrence of failures whereas recovery provides continuity of the system in cases where failures are bound to happen (Mukwevho & Celik, 2018; Adjei et al., 2026).

There are also performance trade-offs that exist between prevention and recovery strategies. The methods of preventive measures usually involve constantly tracking, predictive analysis, and intelligent handling of systems, which can add extra computational load and resource usage (Pentyala, 2024; Jha, 2025). Such monitoring systems have the potential to enhance the reliability of the system, but they can also create more operational complexity in large-scale distributed infrastructures (Abdi et al., 2024; Shandilya et al., 2024). Instead, recovery techniques typically involve extra infrastructure facilities like backup servers, redundant storage systems, and replication strategies to provide the ability to restore the system fast (Achary, 2021; Welsh & Benkhelifa, 2020). Though these mechanisms provide service continuity in case of failures, they can raise the storage demands and the costs of infrastructure in large deployments to the cloud (Nikolovski et al., 2025; Gupta, 2024).

The other factor in the comparison of these approaches is the cost implication and scalability issues. The prevention mechanisms usually involve investment in monitoring infrastructure, AI-driven analytics solutions, and smart coordination systems that constantly examine the performance of the system (Ahmed et al., 2025; Pentyala, 2024). Although these systems enhance the early detection of faults, they might need some advanced computing resources and data management systems to be effective in large-scale settings (Gupta, 2024; Andreyev, 2025). Architectures based on redundancy, backup infrastructures, and disaster recovery mechanisms are, however, required to revive the services in case failures have occurred (Rana & Giri, 2026; Sahar Saeed et al., 2025). Since cloud infrastructures keep growing and adopting new and emerging technologies like IoT and edge computing, it is important to integrate proactive prevention with effective recovery strategies to ensure reliable and resilient cloud computing environments (Ibitoye, 2018; Lampropoulos et al., 2024; Obidiagha & Antwi, 2026) as they grow and become more integrated with new technologies.

Table 7 Comparative Analysis of Prevention vs Recovery Techniques

Aspect	Prevention Techniques	Recovery Techniques
Goal	Avoid system failures through proactive monitoring and anomaly detection	Restore system functionality after failures occur
Time of action	Before failure occurs	After failure occurs
Cost	Monitoring infrastructure and analytics overhead	Backup infrastructure and recovery overhead
Effectiveness	Reduces probability of disruptions	Restores services and minimizes downtime
Resource usage	Continuous monitoring and predictive analysis	Redundant systems and backup resources
Scalability challenge	Managing large monitoring datasets and analytics workloads	Managing replication, redundancy, and disaster recovery infrastructure

7. Research Challenges and Future Directions

Although there have been considerable advancements in the study of cloud reliability, there are still several issues in devising scalable and robust facilities to manage failures. Among these issues, one of the most significant concerns is the scaling of fault-tolerant systems in the large-scale distributed cloud computing system. With the growth of cloud platforms to serve billions of connected things and applications with heavy data requirements, it is further complicated

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